



E-Mail: nationalbgudiaautonomous@gmail.com

☎ 080-26674441

THE NATIONAL COLLEGE BASAVANAGUDI, BENGALURU- 560 004

AUTONOMOUS NAAC Accredited 'A' Grade

Affiliated to the Bengaluru City University Website: www.ncbgudi.com

B S C ANIMATION AND VFX Semester 1 - Year 2024-25							
	Subjects					Marks	
Code	Course	L	P	Exam	IA	Total	Credits
BSCANIC1-DA-T1	Digital Art	3	-	80	20	100	3
BSCANIC1-GD-T2	Introduction to Graphic Design & Creating Illustrations	3	-	80	20	100	3
BSCANIC1-FA-T3	Fundamentals of Art	3	-	80	20	100	3
BSCANIC1-DA-P1	Digital Art PAPER- 2	-	4	40	10	50	2
BSCANIC1-GD-P2	Introduction to Graphic Design & Creating Illustrations PAPER-2	-	4	40	10	50	2
BSCANIC1-FA-P3	Fundamentals of Art PAPER-2	-	4	40	10	50	2
BSCANIC1-LK1	English - 1	4	-	80	20	100	3
BSCANIC1-LX1	Kannada-1,Sanskrit-1	4	-	80	20	100	3
EVS	Environmental Studies	2	-	40	10	50	2
		19	12	560	140	700	23

B S C ANIMATION AND VFX Semester 2 - Year 2024-25							
	Subjects					Marks	
Code	Course	L	P	Exam	IA	Total	Credits

BSCANIC2-FP-T1	Fundamentals of Pre-Production	3	-	80	20	100	3
BSCANIC2-VE-T2	Introduction to Video Editing	3	-	80	20	100	3
BSCANIC2-CP-T3	Cinematography and Photography Concepts	3	-	80	20	100	3
BSCANIC2-FP-P1	Fundamentals of Pre-Production PAPER- 2	-	4	40	10	50	2
BSCANIC2-VE-P2	Introduction to Video Editing PAPER-2	-	4	40	10	50	2
BSCANIC2-CP-P3	Cinematography and Photography Concepts PAPER-2	-	4	40	10	50	2
BSCANIC2-LK2	English - 2	4	-	80	20	100	3
BSCANIC2-LX2	Kannada-2,Sanskrit-2	4	-	80	20	100	3
IC	The Constitution of India	2	-	40	10	50	2
		19	12	560	140	700	23

B S C ANIMATION AND VFX Semester 3 - Year 2024-25							
	Subjects					Marks	
Code	Course	L	P	Exam	IA	Total	Credits
BSCANIC3-DC-T1	Digital Compositing	3	-	80	20	100	3
BSCANIC3-IM-T2	Introduction to 3D Modelling	3	-	80	20	100	3
BSCANIC3-AN-T3	2D Animation	3	-	80	20	100	3
BSCANIC3-DC-P1	Digital Compositing PAPER- 2	-	4	40	10	50	2
BSCANIC3-IME-P2	Introduction to 3D Modelling PAPER-2	-	4	40	10	50	2
BSCANIC3-ANP-P3	2D Animation PAPER-2	-	4	40	10	50	2
BSCANIC3-LK3	English - 3	4	-	80	20	100	3

BSCANIC3-LX3	Kannada-3,Sanskrit-3	4	-	80	20	100	3
UNI SBJ	University Subjects	2	-	40	10	50	2
		19	12	560	140	700	23

B S C ANIMATION AND VFX Semester 4 - Year 2024-25							
	Subjects					Marks	
Code	Course	L	P	Exam	IA	Total	Credits
BSCANIC4-TS-T1	Texturing and Shading 3D Models	3	-	80	20	100	3
BSCANIC4-MM-T2	Match Moving & Camera Tracking	3	-	80	20	100	3
BSCANIC4-LR-T3	Lighting and Rendering	3	-	80	20	100	3
BSCANIC4-TS-P1	Texturing and Shading 3D Models PAPER- 2	-	4	40	10	50	2
BSCANIC4-MM-P2	Match Moving & Camera Tracking PAPER-2	-	4	40	10	50	2
BSCANIC4-LR-P3	Match Moving & Camera Tracking PAPER-2	-	4	40	10	50	2
BSCANIC4-LK4	English - 4	4	-	80	20	100	3
BSCANIC4-LX4	Kannada-4,Sanskrit-4	4	-	80	20	100	3
UNI SBJ	University Subjects	2	-	40	10	50	2
		19	12	560	140	700	23

B S C ANIMATION AND VFX Semester 5 - Year 2024-25							
	Subjects					Marks	
Code	Course	L	P	Exam	IA	Total	Credits
BSCANIC5-RA-T1	3D Rigging and Animation	3	-	80	20	100	3
BSCANIC5-IN-T2	Internship					200	6
BSCANIC5-MG-T3	Motion Graphics	3	-	80	20	100	3
BSCANIC5-PDM-P1	Portfolio Development (Motion Graphics)	-	4	40	10	50	2

BSCANIC5-PDG-P2	Portfolio Development (Graphic Design)	-	4	40	10	50	2
UNI SBJ	University Subjects	2	-	40	10	50	2
		9	8	280	70	550	18

B S C ANIMATION AND VFX Semester 6 - Year 2024-25							
	Subjects					Marks	
Code	Course	L	P	Exam	IA	Total	Credits
BSCANIC6-PD-T1	Particle Dynamics	3	-	80	20	100	3
BSCANIC6-IR-T2	Internship					200	6
BSCANIC6-CM-T3	3D Compositing	3	-	80	20	100	3
BSCANIC6-PD3-P1	Portfolio Development 3D Modelling	-	4	40	10	50	2
BSCANIC6-PDR-P2	Portfolio Development Rigging & Animation	-	4	40	10	50	2
BSCANIC6-PDV-P3	Portfolio Development Project Work on VFX		4	40	10	50	2
UNI SBJ	University Subjects	2	-	40	10	50	2
		9	12	320	80	550	20

Detailed Syllabus for B.Sc Animation and VFX

Year: 1	Semester: 1	BSCANIC1-LK1	English - 1	Credits: 3
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Objectives:

1. To introduce students to the theory, fundamentals and tools of communication and to develop in them vital communication skills that is integral to personal, social and professional interactions.
2. Inculcate abilities needed to students such as: ability to share thoughts, emotions and ideas through various means of communication: both verbal and non-verbal. In the context of rapid globalization and increasing recognition of social and cultural pluralities, the significance of clear and effective communication has substantially enhanced
3. Fulfil the above state goals through an interactive mode of teaching-learning process and by focusing on various dimensions of communication skills: language of

communication, various speaking skills such as personal communication, social interactions and communication in professional situations such as interviews, group discussions and office environments, important reading skills as well as writing skills such as report writing, note-taking etc.

4. While, to an extent, the art of communication is natural to all living beings, in today's world of complexities, it has also acquired some elements of science. It is hoped that after studying this course, students will find a difference in their personal and professional interactions.

SYLLABUS:

As stipulated by the BENGALURU CITY UNIVERSITY

Year: 1	Semester: 1	BSCANIC1-LX1	Kannada-1,Sanskrit-1	Credits: 3
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Objectives: To impart working knowledge of multi-lingual ability to students as part of projecting the composite culture of our country envisaged under Article 351 of the Indian Constitution and for strengthening national integration, Indian languages like Kannada, Tamil, Telugu, Malayalam and Hindi are offered. Students can choose any one of the above languages for study.

SYLLABUS

As stipulated by the BENGALURU CITY UNIVERSITY

Year: 1	Semester: 1	BSCANIC1-DA-T1	Digital Art	Credits: 3:2
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Objectives:

To familiarize the students with the tools used in Adobe Photoshop and its application for typography, image editing, digital painting etc. for print and digital media.

SYLLABUS

UNIT I: Getting started with Adobe Photoshop CC – Overview of Adobe Photoshop CC, Features, Overview of tools used, Importance of Adobe Photoshop CC

UNIT II: Working with Typography – Creating Typographies, Choosing the right font and color

UNIT III: Working with Layers and Images – Cropping a Photo, Resizing Images, Basics of Layers, Creating Layers for Print and Digital Media, Aligning Images with Multiple Layers, Merging Layer Techniques

UNIT IV: Working with Filters – Photoshop Filters, Smart Filters, Common Features of Filters

UNIT V: Digital Painting – working with brush tool, importance of using colors

UNIT VI: Masking and File Formats – Introduction to Mask, Creating Vector and Layer Masks, Essential File Formats, Choosing the Right format for Print and Digital Media

REFERENCES

1. Basic Drawing Techniques by Richard Box Pub: Winsor & Newton, (U.S.A)

2. Still Life by Sanjay Shelar, JyotsanaPrakashan(India).Pub.
3. Drawing and Anatomy by Victor Perard , Kingsport Press Pub(U.K).

Year: 1	Semester: 1	BSCANIC1-GD-T2	Introduction to Graphic Design & Creating Illustrations	Credits: 3:2
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Objectives:

To learn the process of graphic design including the principles and elements of design, different color types and visualization techniques that are required in making a good design using Adobe Illustrator.

SYLLABUS

UNIT I: Design Basics - Introduction to Design, History of Design, Content and Form, Rules for making Good Design, Graphic Design Process, Creating Effective Layouts

UNIT II: Principles and Elements of Design - Principles of Design, Elements of Design, Design Problems to Avoid

UNIT III: Understanding Colors – Introduction, Color Wheel, Primary and Secondary Colors, Black and White, Tertiary Colors, Warm and Cool Colors, Color Models, Hue, Saturation, Value, Transparent and Opaque Colors, Indexed Color, True color

UNIT IV: Graphical Aspects of Design and Computer Graphics – Aspects of Design, Introduction to Computer Graphics, Effects

UNIT V: Visualization - Visualization and Imagery Techniques, Direct And Indirect Approach, Thinking in various points of view

UNIT VI: Resolution and File Formats – Understanding resolution, image resolution, device resolution, printer resolution, interpolated resolution, digital image, file size and bit depth, scanning images, graphic file format.

UNIT VII: Understanding and working with Graphics and Web Images – Understanding Graphic images, types of graphic program, file formats for web, colors and text for the web.

REFERENCES

- 1) Graphic Design for the 21st century - by Charlotte Fiell, Peter Fiell
- 2)What is Graphic Design - by Quentin Newark
- 3)Graphic Design - by AarisSherin
- 4)Adobe Photoshop - by Bittu Kumar
- 5)Adobe Illustrator - by Code Kart
- 6) Adobe Photoshop - by Code Kart
- 7)Adobe illustrator - by Schwabe

Year: 1	Semester: 1	BSCANIC1-FA-T3	Fundamentals of Art	Credits: 3:2
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Objectives:

To understand the basics of Art such as light, color and perspectives from various elements of nature and learn fundamentals of character sketching.

SYLLABUS

UNIT I: Outdoor studies – observing forms and shapes, Adding Tones, Shading with Texture Quality, Contour Drawing, Cinematography Drawing, Creating Rough sketches, Shadows, Overlapping Objects, Time and Season, Layers and Character Staging

UNIT II: Lighting and Color Concepts – Understanding Lights, Understanding proper light source, lighting and shading, drawing groups of basic shapes, Primary, secondary, tertiary and quaternary color, warm and cool color applications, greyscale keys

UNIT III: Perspective study – understanding perspective, one, two, three and four point perspective

UNIT IV: Character Design Concepts – Role of a character designer, designing character, creating character drawings, character based on age, creating four legged animal, add-on features

REFERENCES

1. Art fundamentals : Color, Light, Composition, Anatomy – 3Dtotal publishing
2. Light for visual artists : Understanding & using visual light in art & design – Richard Yot
3. Bridgman's complete guide to drawing from life – George B Bridgman
4. How to draw : Drawing and sketching objects and environments

Year: 1	Semester: 1	EVS	Environmental Studies	Credits: 2
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To develop the sense of awareness among the students about environment and its various problems.
To help the students in realizing the inter-relationship between man and environment.

SYLLABUS

As stipulated by the BENGALURU CITY UNIVERSITY

Year: 1	Semester: 2	AECC 1B	BSCANIC2-LK2	Credits: 3
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SYLLABUS

As stipulated by the BENGALURU CITY UNIVERSITY

Year: 1	Semester: 2	BSCANIC1-LX1	Kannada-1,Sanskrit-1	Credits: 3
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As stipulated by the BENGALURU CITY UNIVERSITY

Year: 1	Semester: 2	BSCANIC2-FP-T1	Fundamentals of Pre-production	Credits: 3:2
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Objectives:

To discuss the basics of storytelling, scriptwriting, composition of a creative outline that is involved in the production of any film or commercial. This course also discusses the nuances of storyboarding for Animatics.

SYLLABUS

UNIT I: Storytelling Techniques - Introduction to Cartoon Animation Writing, Introduction to Storytelling, Adaptation, Telling a Good Story, Structuring a Story, Writing a Story

UNIT II: Scriptwriting - Introduction to Scriptwriting, Tips to Construct a Story, Scriptwriting Format and Elements, Modern Ways to Write Scripts, Creating Scripts for Animation Films, Creating Scripts for Commercials

UNIT III: Introduction of Animatics – About Animatics, Difference between storyboards and Animatics, Application areas of Animatics

UNIT IV: Enhancing storyboards for Animatics – Scanning storyboards, enhancing lines, using colors, coloring background and characters, applying hues and tones, using filters

Reference Books:

1. The Art of Layout and Storyboarding (Author: Mark Byrne)
2. Shot by shot (Author : Steven D Katz)
3. Animation Script to Screen (Author: Shamus Culhane)

Year: 1	Semester: 2	BSCANIC2-VE-T2	Introduction to Video Editing	Credits: 3:2
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Objectives:

To familiarize students about audio and video editing using Adobe Audition and Premiere Pro interface and to get started with editing, working with clips and adding different effects.

SYLLABUS

UNIT I: Introduction to Adobe Audition - Fundamentals of Sound, Understanding Adobe Audition Interface, Adjusting Audio Preferences, Creating and Opening Files

UNIT II: Working with Audio Editing - Editing Audio Clips, Saving and Exporting File, Batch Process, Converting Sample Type

UNIT III: Editing Video - Basic Workflow, Adding, Rearranging, and Working with Clips in a Sequence, Rendering and Previewing Sequences, Overview of Audio and Audio Track Mixer, Working with Clips, Channels, and Tracks, About Effects - Applying, Removing, Finding, and Organizing Effects, Applying Audio Effects and Transitions

REFERENCES

1. Sound and Recording: Francis Rumsey, Tim McCormick
2. Modern Recording Techniques: David Miles Huber
3. Audio post production for film and Video: Jay Rose
4. Non-linear Editing: Bryce Button
5. Notes on Digital Film Editing: Gerhard Schumm

Year: 1	Semester: 2	BSCANIC2-CP-T3	Cinematography and Photography Concepts	Credits: 3:2
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Objectives:

On completion of the course, students will gain knowledge about camera, type of lighting, composition techniques for photography and sensors involved in photography and cinematography

SYLLABUS

UNIT I: Photography Concepts – Camera, Lighting, Composition Techniques

UNIT II: Cinematography Concepts – Cameras and sensors, lighting, exposure, camera movement

REFERENCES

1. Cinema Studies, The Key Concepts (3rd Ed.): Susan Haywood, London: Routledge
2. How to Read A Film: James Monaco. New York: OUP, 2000
3. Genre and Contemporary Hollywood: Steve Neal (Ed.). London: BFI, 2002

Year: 1	Semester: 2	IC	The Constitution of India	Credits: 2
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SYLLABUS

As stipulated by the BENGALURU CITY UNIVERSITY

Code:	Year: 2	Semester: 3	BSCANIC3-LK3	English - 3	Credits: 3
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As stipulated by the BENGALURU CITY UNIVERSITY

Code:	Year: 2	Semester: 3	BSCANIC3-LX3	Kannada-3,Sanskrit-3	Credits: 3
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As stipulated by the BENGALURU CITY UNIVERSITY

Code:	Year: 2	Semester: 3	BSCANIC3-IM-T2	Introduction to 3D Modelling	Credits: 3:2
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Objectives:

To understand the basics of 3D modelling such as 3D axes, projections, workspace and coordinate system and understand the processes involved from modelling up to post production.

SYLLABUS

UNIT I: Overview of 3D - Understanding 3D, 3D Projection, 3D Geometry, 3D Workspace

UNIT II: 3D Modeling to Rendering Process – Modeling, Materials and Textures, Lighting, Rigging, Animation, Rendering

UNIT III: 3D Animation Environment and Post-Production – Views and Projections

Reference Books:

1. Animation: From Pencil to Pixels by Tony White
2. Animator's Survival Kit – Richard Williams
3. The Illusion of Life – Frank Thomas &OlieJohnstan

4. Animation Script to Screen (Author: Shamus Culhane)
5. Animation Writing and Development: From Script Development to Pitch By Jean Ann Wright
6. The History of Moviemaking: Animation and Live-Action, from Silent to Sound, Black-AndWhite to Color (Voyages of Discovery), Scholastic Trade

Code:	Year: 2	Semester: 3	BSCANIC3-AN-T3	2D Animation	Credits: 3:2
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Objectives:

Understanding the concepts and principles involved in classical animation and 2D Digital Animation concepts using Animate CC.

SYLLABUS

UNIT I: Introduction to Principles of Animation – Stretch and Squash, Timing and Spacing, Ease in and ease out, Arcs, follow through and overlapping, staging, anticipation, exaggeration, Animation Principles in Animate

UNIT II: Introduction to Animate CC – About Animate CC, Workspace, manage windows and panels, using the stage

UNIT III: Working with timeline, layers and keyframes - Understanding the Timeline, Changing the Appearance of the Timeline, Changing the Display Frames in the Timeline, Onion Skinning, Moving the Playhead, Using layers and keyframes

UNIT IV: Shapes and Symbols – Creating basic shapes, stroke and fill, patterns, types of symbols, editing symbols, using library panels

UNIT V: Creating Animations – Understanding Motion tween, classic tween, shape tween, suing sound in animate CC

REFERENCES

- 1) The Animator Survival Kit by Richards Williams
- 2) Basic Drawing Techniques by Richards Box
- 3) Drawing and Anatomy by Victor Perard
- 4) Sketching by Pratap Mulik
- 5) Human Anatomy for Artist by Eliot Goldfinger

Code:	Year: 2	Semester: 3	BSCANIC3-DC-T1	Digital Compositing	Credits: 3:2
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Objectives:

At the end of this course, student will be able to define various types of compositing, create a composition for a project with appropriate parameters, learn to manage layers, practice animation, perform effects and more using Adobe After Effects.

SYLLABUS

UNIT I: Getting started with Adobe After Effects CC - Introduction to After Effects, Setup and Installation, System Requirements, General workflow in After Effects CC, Menu Bar, Tool Bar, and Panels, Dynamic Links, Working with Other Applications, Projects and Composition

UNIT II: Adding Various Digital Assets and Previewing the Composition - Importing Digital Assets, Interpreting Footages, Details of Files, Switches and Columns, Working on Digital Assets, Arranging and Managing Layers, Previewing the Composition, Region of Interest, Snapshots

UNIT III: Creating Animation Effects - Animation Basics, Keyframes, Expressions, Graph Editor, Motion Paths, Motion Blur, Animating with Puppet Tools, Time-based Effects

UNIT IV: Text Animation and Animation Effects - Applying Effects Using Preset, Mask, Color Keying, Working with Green Screen Footages, Blending Modes, Text Animation, Particles

UNIT V: 3D in After Effects, Previewing and Rendering Output

REFERENCES

1. Editing Digital Video: The Complete Creative and Technical Guide by Robert Goodman (McGraw-Hill)
 2. Digital compositing for film & video by Steve Wright
- Professional digital compositing: Essential Tools and Techniques by Lee Lanier, Pub. - Sybex.

Code:	Year: 2	Semester: 4	BSCANIC4-LK4	English - 4	Credits: 3
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As stipulated by the BENGALURU CITY UNIVERSITY

Code:	Year: 2	Semester: 4	BSCANIC4-LX4	Kannada-4,Sanskrit-4	Credits: 3
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As stipulated by the BENGALURU CITY UNIVERSITY

Code:	Year: 2	Semester: 4	BSCANIC4-TS-T1	Texturing and Shading 3D Models	Credits: 3:2
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Objectives: On completion of this topic, students will be able to understand the concepts in shading and texturing 3D Models using Maya for various models and characters.

SYLLABUS

UNIT I: Introduction to Shading and Texturing - Materialistic Properties of Objects, Shader/Material, textures, UVW Co-ordinate System, Maya Hypershade, Working with Maya Shading Nodes - Blinn, Phong,an Lambert Shading Nodes

UNIT II: Working with 2D and 3D Textures - Procedural Textures - An Introduction, Difference between Bitmap and Procedural Textures, Working with 2D Procedural Textures, Working with 3D Procedural Textures

UNIT III: Basics of UV Mapping – Bump, Normal and Displacement Maps

UNIT IV: Layered Shader and Layered Texture - Understanding the Layered Shade, Applying Layer Shaders, Understanding the Layered Texture, Working with Layered Textures

UNIT V: Introduction to UV Unwrap – BG – UVs, UV Editor, UV Mapping Techniques

UNIT VI: Unwrapping and texturing a character – Skin shaders, Diffuse Maps, Specular Maps, creating Bump Maps, Texturing body and apparel

UNIT V: Fundamentals of Hypershade and Utilities

REFERENCES

1. Principles of Form and Design by Wucius Wong John Wiley & Sons, New York
2. Digital compositing for film & video by Steve Wright
3. Professional digital compositing: Essential Tools and Techniques by Lee Lanier, Pub. - Sybex.

Code:	Year: 2	Semester: 4	BSCANIC4-MM-T2	Match Moving and Camera Tracking	Credits: 3:2
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Objectives:

To understand about match moving, a cinematic technique that allows the insertion of computer graphics into live-action footage with correct position, scale, orientation, and motion relative to the photographed objects in the shot.

SYLLABUS

UNIT I: Tracker Node - Need for Tracker Node, Viewer Tools, Positioning Track Anchors, Automatic Keyframe Tracking, Applying Track Data

UNIT II: Planar Tracker - Connecting the Planar Tracker Node, Planar Tracking, Altering Tracking Results, Adding Image to Planar Surface

UNIT III: Camera Tracking - Steps to Connect the Camera Tracker Node, Masking, Multi-View Scripts, Camera Parameters, Sequence Tracking

UNIT IV: Still Mode - Tracking Still Frames, Viewing Reference Frames and Track Data, Frame Sets, Troubleshooting

UNIT V: User Tracks & Solving Camera- Adding User Tracks, Tracking Methods, Tracking Assists, Tracking a Scene Manually, Assigning 3D Survey Points, Viewing Solving Data, Troubleshooting

Reference Books:

1. Matchmoving: The Invisible Art of Camera Tracking - Tim Dobbert, Cdr edition, Sybex, 2005
2. The Art and Technique of Matchmoving: Solutions for the VFX Artist - Erica Hornung, 1 edition, Routledge, 2010
3. The Filmmaker's Guide to Visual Effects: The Art and Techniques of VFX for Directors, Producers, Editors and Cinematographers - EranDinur, 1 edition, Routledge, 2017
4. [digital] Visual Effects and Compositing - Jon Gress, 1st Edition, New Riders; 2014

Code:	Year: 2	Semester: 4	BSCANIC4-LR-T3	Lighting and Rendering	Credits: 3:2
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Objectives:

To learn the Fundamentals and principles of Lighting of a 3D model and publishing it with rendering.

SYLLABUS

UNIT I: Fundamentals of Standard Lights and Shadows – Basic properties of Lights, Types of Lights in Maya, Types of Shadows

UNIT II: Studio Lighting and three-point lighting.-Principles of three-point lighting, planning the setup, lighting the character, shadows, important tips on light setup

UNIT III: Day Lighting- Exterior Model and fill lights – Creating a day light setup on a background, using fill lights, using rim and Hi Lights for the background, Character Lighting setup based on the background, Pre-compositing character and background

UNIT IV: Night light -Interior light and passes – Lighting the background, creating a Moonlight setup on an Interior Background

UNIT V: Image Based lighting – Introduction to HDRI, Lights and Shadows in the Live Plate, Adjusting the camera as per the Live Plate, Shading and Lighting for Cyborg Character, Color correction

REFERENCES

1. Introducing Autodesk Maya 2014 Basic Guide By DariushDerakshani
2. Color And Light : A Guide For The Realist Painter (James Gurney Art)
3. Digital Lighting And Rendering – 1st And 2nd Edition. – By JermyBim

Code:	Year: 3	Semester: 5	BSCANIC5- RA-T1	3D Rigging and Animation	Credits: 3:2
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Objectives:

To understand the fundamentals of Rigging and Animation required to a good 3D Animation model that can be used for games, movies etc.

SYLLABUS

UNIT I: Introduction to Rigging – Concept of Rigging in Maya, Relation between Objects, Designing a chair Rig

UNIT II: Working with Constraints and Set Driven Key – Set Driven Key, Creating a Functional Rig Setup of a Cuckoo Clock

UNIT III: Mechanical Rig (Usage of constraints, utilities, SDK) – Aim Constraint, Developing a Functional Mechanical Rig

UNIT IV: Deformers - Lattice, Wrap, Cluster – What are Deformers?, Lattice Deformer, Wrap Deformers, Clusters

UNIT V: Working with Sculpt, Jiggle, and Wire Tool Deformer – Sculpt Deformer, Jiggle Deformer, Wire tool deformer

UNIT VI: Introduction to Character Rigging – Concept of Character Rigging, Introduction to joints, Joint creation process, develop a join system for a character

UNIT VII: Introduction to Timeline and Keyframe Animation – Animation Interface, Animation Preferences, Understanding Key frame Animation

UNIT VIII: Bouncing Ball – Introduction to Dope Sheet, Understanding the graph editor, tangents, Bouncing ball assignment, gravity

UNIT IX: Pendulum Exercise – Pendulum Animation Fundamentals, Arc of Motion, Animating the pendulum, Secondary Action (Drag)

UNIT X: Wave Exercise – working on wave animation

UNIT XI: Working with weights – Sack Exercise – Explain the basics of Weight, Adding emotions, Flour Sack – Jumping Actions

UNIT XII: Creating Strong Poses – Gesture Drawings – An introduction, Line of Action, Negative Space Versus Positive Space, Silhouettes, Creating a pose using line of action

REFERENCES

- 1) An Introduction to Rigging in the Entertainment Industry (Applications & Techniques)
- 2) Art of Rigging by George Biddlecombe, Pub.- Dover Publications by Chris Higs, Pub.- Entertainment Technology Press Ltd.

Code:	Year: 3	Semester: 5	BSCANIC5- IN-T2	Internship	Credits:3:3
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The student is expected to work in any company for a minimum period of 6 weeks. He is supposed to work on the assignments given by the company and submit the same as per the schedule given. He is supposed to submit a report from the company on successful completion. Alternatively if he not able to get any company internship, the college can assign a simulated project to the student. On submission, the same can be evaluated by the respective lecturer.

Code:	Year: 3	Semester: 5	BSCANIC5- MG-T3	Motion Graphics	Credits: 3:2
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Objectives:

On completion of the course, students will be able to understand the basics of motion graphics, special effects using Adobe After Effects and applying effects in Adobe After Effects.

SYLLABUS

UNIT I: Introduction to Motion Graphics - Pal and NTSC formats, Frame rate, Adobe After effect interface, Panels ,composition, composition setting, nested composition, pre composition, preferences, importing files, transform properties, key farming, basic animation using transform properties, motion blur, interpolation, graph, ,blending modes

UNIT II: Case study of motion graphics video, Text tool, Parenting, Masking, types of masking tool, stroke application of masking, creation of motion graphics video(explainer video)

UNIT III: Making Basic animation using keyfarme, Creating Basic motion graphics video, Creating rotoscopy video, Creating explainer video, Creating cinematic effects, Creating chroma video, Creating tracking video

Reference Books:

1. Adobe After Effects Classroom in a Book (2021 release) by [Lisa Fridsma](#)
2. Creating Motion Graphics with After Effects: Essential and Advanced Techniques, 5th Edition, Version CS5 by Chris Meyer and Trish Meyer
3. Adobe After Effects CS6 Digital Classroom Book by Jerron Smit.
4. Creating Motion Graphics with After Effects by Chris Meyer and Trish Meyer

Code:	Year: 3	Semester: 5	BSCANIC5-PDM-P1	Portfolio Development (Motion Graphics)	Credits: 2
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Portfolio Development in Motion Graphics**Objective:**

To showcase the skills learnt in the Motion Graphics Designing by creating a portfolio on any topic of their choice. One of the lecturer will be assigned as mentor who will guide throughout the period of portfolio development. The portfolio should be submitted in the format mentioned by the lecturer in the stipulated time frame.

Code:	Year: 3	Semester: 5	BSCANIC5-PDG-P2	Portfolio Development (Graphic Design)	Credits: 2
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Portfolio Development in Graphic Design**Objective:**

To showcase the skills learnt in the Graphic designing Designing by creating a portfolio on any topic of their choice. One of the lecturer will be assigned as mentor who will guide throughout the period of portfolio development. The portfolio should be submitted in the format mentioned by the lecturer in the stipulated time frame.

Code:	Year: 3	Semester: 6	BSCANIC6-PD-T1	Particle Dynamics	Credits: 3:2
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Objectives:

1. To understand how to paint on canvas using various types of brushes and paint strokes.
2. To understand the concept of nCloth system
3. To create dynamic simulations for clothes
4. To describes working with Maya Fur

SYLLABUS**UNIT I:**

Visual Effects- Description- Types- Particles – Analysis- Size- Sand Effects – Smoke Effects- Fire Effects – Cloud Effects – Snow Effects

UNIT II:

Fluid Effects-Coloring- designing Clouds Background – Designing Fog Effects – Explosion Effects– Fire Effects with flames - Space Effects and designs- Designing Thick Smoke

UNIT III:

Maya nDynamics – Clothing and Hair

Creating Clothing for Character, Create Passive Collider, Dynamic Properties, Creating nCloth and nParticles Interaction, Hair Overview, Hair Creation, Adding Hair to Character, XGen in Maya, Work with Clump Modifiers, Collisions, Dynamic Properties

UNIT IV:

Designing Paint Effects – Coloring paints- Designing Trees and green effects – Designing Weather and seasons –Effects on seasons- Designing Glass image – Designing Different glass reflection- Designing Glow Effects – Liquid Effects and reflection design

UNIT V:

Designing Special Effects – Designing effects of Hair and shape – Designing Fur EffectsDesigning Clothes and effects

Reference Books:

1. Autodesk Maya 2018: A Comprehensive Guide Paperback - by Tickoo Sham, November 1, 2017, BPB Publications
2. Maya Studio Projects: Dynamics - Todd Palamar, Nov 2, 2009, Sybex
3. The Art of Fluid Animation - Jos Star, Oct 30, 2015, A K Peters/CRC Press

Code:	Year: 3	Semester: 6	BSCANIC6- CM-T3	3D Compositing	Credits: 3:2
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Objectives:

To understand general workflow of Nuke, the basic features of Nuke window, the functions of tools available in the Nuke Toolbar, working of key frames in the Dope sheet and the concept of nodes and how to work with it.

SYLLABUS

UNIT I: Getting Started with Nuke - Components of the Graphic User Interface, Understanding Workflow, Working with Nodes, Working with Nuke Window, Working with Nuke Tool Bar, Working with Nuke Menu Bar, Understanding the Viewer, Understanding Framecycler

UNIT II: Understanding the Interface - Basic Composite Working with Process Trees, Creating a Simple Process Tree, Merging Images, Animating Parameters, Using Curve Editor, Customizing the Interface

UNIT III: Understanding Nodes - Touring the Interface with a Basic Composite Working with Nodes, Selecting Nodes, Renaming Nodes, Inserting and Manipulating Nodes in the Tree, Changing Properties, Rendering, Touring the Interface with a Basic Composite Editing Nodes, Cloning Nodes, Understanding Merge Node, Creating Animation with Keyframes

UNIT IV: Compositing final renders - How to use multi render system.

Reference Books:

1. The Foundry Nuke X 7 for Compositors Paperback-by Prof. Sham Tickoo Purdue Univ. (Author), CADCIM Technologies (Author), Edition – 1st (June 20, 2013), Publication: CADCIM Technologies
2. Digital Compositing for Film and Video: Production Workflows and Techniques- by Steve Wright (Author), Edition -4th, Publication: Routledge

Code:	Year: 3	Semester: 6	BSCANIC6- IR-T2	Internship	Credits: 3:3
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The student is expected to work in any company for a period of 6 weeks. He is supposed to work on the assignments given by the company and submit the same as per the schedule given. He is supposed to submit a report from the company on successful completion. Alternatively if he not able to get any

company internship, the college can assign a simulated project to the student. On submission, the same can be evaluated by the respective lecturer.

Code:	Year: 3	Semester: 6	BSCANIC6- PD3-P1	Portfolio Development 3D Modelling	Credits: 2
Code:	Year: 3	Semester: 6	BSCANIC6- PDR-P2	Portfolio Development Rigging & Animation	Credits: 2
Code:	Year: 3	Semester: 6	BSCANIC6- PDV-P3	Portfolio Development Project Work on VFX	Credits: 2

Objective:

To showcase the skills learnt in the 3D Modelling / Rigging & Animation / VFX by creating a portfolio on any topic of their choice. One of the lecturer will be assigned as mentor who will guide throughout the period of portfolio development. The portfolio should be submitted in the format mentioned by the lecturer in the stipulated time frame.